■ CampusCarbon

Stage 2 Submission

SDG 13 · Climate Action	SDG 4 · Quality Education	SDG 11 · Sustainable Cities

Product Overview

CampusCarbon is an AI-powered carbon footprint tracker built specifically for college students. Students log their daily habits — meals, commute, and energy usage — and instantly see their personal CO2 emissions score, compare with peers on a campus leaderboard, and receive AI-generated nudges to make greener choices. Built for the realities of student life in India and Southeast Asia: mess food, hostel rooms, college buses, and auto-rickshaws.

The Problem

2.55 tonnes	50%+	0
avg annual CO2 per student (InnsPub 2022)	students have LOW carbon awareness (Academia 2025)	tools built specifically for campus life

Existing tools like Joro require credit card linkage (irrelevant for most students). The WWF Footprint Calculator is a one-time quiz with no ongoing engagement. No tool exists that is built around the daily rhythm of campus life.

The Solution — Four Core Features

■■ Daily Carbon Logger

A 3-tap logging flow for meal, commute, and energy. Each option shows CO2 in real time. Students can add custom options with their own CO2 values.

■ Carbon Score Dashboard

A visual ring-based score, 7-day trend chart, weekly average, CO2 saved vs average, streak counter, and campus rank — all on one screen.

■ Campus Leaderboard

All registered students ranked by average daily CO2. Peer competition drives engagement — the same mechanic that makes Duolingo sticky, applied to sustainability.

■ AI Coach (Groq / LLaMA 3.3 70B)

Conversational AI coach gives personalised, campus-specific advice referencing mess food, autos, and hostel energy use. Knows the student's live stats.

Technology Stack

Layer	Technology
Frontend	React 18 + Vite
Backend	Node.js + Express
Database	MongoDB Atlas (persistent, cloud-hosted)
AI	Groq API — LLaMA 3.3 70B Versatile (free tier)

Auth	bcrypt password hashing
Styling	Custom CSS + Outfit font

How This Maps to the Stage 1 Business Case

Problem Validation

Stage 1 identified students emit 2.55t CO2 annually with 50%+ unaware of their footprint. CampusCarbon directly bridges this awareness gap with a daily habit loop backed by real data.

Target User

Stage 1 defined urban college students aged 18-24 in India and Southeast Asia. Every feature reflects this: Indian transport modes (auto, Ola, college bus), Indian meal types, hostel energy use.

SDG Alignment

SDG 13 (Climate Action): each log creates measurable CO2 data. SDG 4 (Education): carbon literacy embedded in daily life. SDG 11 (Sustainable Cities): campuses as micro-sustainability communities.

Revenue Model

Stage 1 proposed a university SaaS licence (Rs. 2-5L/year) for ESG dashboards. The MVP builds the student-facing data layer that makes the institutional dashboard valuable for accreditation.

Go-To-Market

Stage 1 identified WhatsApp groups and sustainability clubs for first 100 users. The leaderboard is the product-led growth engine — one student joins, peers compete to rank higher.

3-Year Vision

Stage 1 projected 1M users across 500 universities, reducing 200,000 tonnes CO2/year. This MVP establishes the core habit loop, auth, persistent logging, and leaderboard that scales there.

Why This Is Defensible

- Behaviour-first design — built around a student's actual daily decisions, not generic carbon categories.
- Network effects — the leaderboard becomes more valuable as more students from the same campus join.
- Institutional lock-in — universities that licence the dashboard promote the app to students, creating a B2B2C flywheel.
- Data moat — longitudinal campus-level emissions data is a unique asset for climate research and ESG reporting.

Built for the hackathon. Designed to scale. ■